

Template Week 1 – Bits & Bytes

Student number:530010

Assignment 1.1: Bits & Bytes intro

What are Bits & Bytes?

A bit (binary digit) is the smallest unit of data in a computer, representing a value of either 0 or 1. A byte consists of 8 bits and is the standard unit used to represent a character of data, such as a letter or symbol.

What is a nibble?

4 bits

What relationship does a nibble have with a hexadecimal value?

A nibble is 4 bits, and a hexadecimal value goes in steps of 4.

Why is it wise to display binary data as hexadecimal values?

Because it provides a compact and human-friendly way of representing large binary numbers.

What kind of relationship does a byte have with a hexadecimal value?

A byte is 8 bits, a hexadecimal digit represents 4 bits. Therefore, 1 byte is 2 hexadecimal digits

An IPv4 subnet is 32-bit, show with a calculation why this is the case.

$4 \text{ octets} \times 8 \text{ bits per octet} = 32 \text{ bits}$

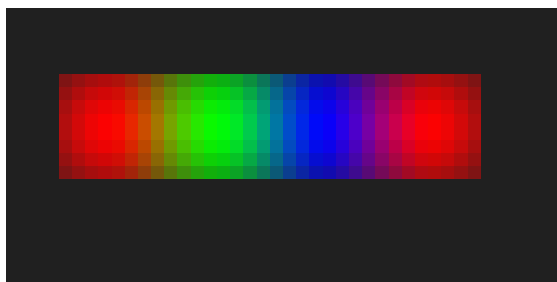
Assignment 1.2: Your favourite color

Hexadecimal color code: #ff0000

Assignment 1.3: Manipulating binary data

Color	Color code hexadecimal (RGB)	Big Endian	Little Endian
RED	#ff0000	ff 00 00	00 00 ff
GREEN	#00ff00	00 ff 00	00 ff 00
BLUE	#0000ff	00 00 ff	ff 00 00
WHITE	#ffffff	ff ff ff	ff ff ff
Favourite (previous assignment)	#ff0000	ff 00 00	00 00 ff

Screenshot modified BMP file in hex editor:



Assignment 1.4: Student number to HEX and Binary

Convert your student number to a hexadecimal number and a binary number.

Explain in detail that the calculation is correct. Use the PowerPoint slides of week 1.

0d 530010

0x 8165A

0b 10000001011001011010

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